

ROUSSEL Victor

Paris Area, France

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Second-year engineering student at CentraleSupélec with a strong interest in applied mathematics, particularly AI, data science and cryptography. Solid theoretical background and regular hands-on modelling of real-world data.

Seeking a 5 to 6 month gap-year internship starting early September 2026.

EDUCATION

CentraleSupélec - Université Paris-Saclay

Paris-Saclay, France

Engineering Degree

Sep 2024 – Jun 2027

- Key courses: Measure Theory & Probability (Hilbert spaces, L^p spaces, conditional expectation, random walks), Partial Differential Equations (distributions, Sobolev spaces, variational formulation), Algebra & Cryptology (lattice reduction, LLL algorithm, elliptic curves, learning with errors), Information Theory (entropy, mutual information, f-divergences), Statistics & Machine Learning, Optimization (convexity, duality, Lagrange multipliers)

Lycée Hoche

Versailles, France

CPGE: MPSI – MP

Sep 2021 – Jun 2024

- Intensive two-year undergraduate program in mathematics and physics, preparing for nationwide competitive entrance exams to top French engineering schools (equivalent to a Bachelor's level in Mathematics).

PROFESSIONAL EXPERIENCE

Lycée Hoche - Éducation Nationale

Versailles, France

Oral Examiner

Sep 2024 – Jun 2026

- Assessed 6 second-year students weekly in 2-hour oral examinations (“khôlles”) in mathematics, preparing tailored exercises for each student's level (see examples) and reporting individual progress to the professor.

Sumécatronic

Magny-les-Hameaux, France

Milling Machine Operator

Jun 2025 – Jul 2025

- Autonomous milling machine operator, producing and quality-controlling series of precision-engineered parts for the aeronautics industry. Gained hands-on experience with quality assurance processes and workshop logistics.

PROJECTS

Lab project: Liver Cancer Classification from Radiomic Features — Henri Mondor Hospital, CentraleSupélec | Gif-sur-Yvette, France

Project team

Jul 2026

- Compared three families of methods (penalised multi-block regression with groupLASSO, random forests, PCA) on an incomplete and imbalanced cohort, each validated over one hundred distinct random splits to assess robustness.
- Best configuration: 93% AUC, 88% accuracy. The most frequently selected variables match the reading criteria described by the oncologists.

Client project: Multi-view Sport Video Synchronisation — GoPro |

Gif-sur-Yvette, France

Synchformer track

Jan 2026 – Jun 2026

- Literature review leading to a comparison of three approaches (human pose detection, audio-visual alignment using transformers, visual similarity matching); fine-tuned a Synchformer-based model on Ego-Exo4D, deployed on DGX A100 through Slurm.
- Mean absolute error of 0.081s on the discrete-event, high-confidence subset; showed the model's confidence score to be calibrated enough to act as a routing signal towards an alternative method.
- Run under professional conditions: regular client meetings, intermediate deliverables and final report.

Course project: Variance Reduction for Monte Carlo Simulation in Financial Mathematics |

Gif-sur-Yvette, France

Project team

Mar 2026

- Implemented and compared eight variance reduction methods across 810 configurations of volatility, dimension and moneyness, scored with $|\text{err}|/\sqrt{\text{time}}$ so that computational cost counts against a method.
- Sobol sequences proved the most efficient method in 46% of configurations; trained a random forest reaching 82% cross-validated accuracy to predict the best method for a given configuration.

Challenge week: Data-Driven Modelling and Control of a UAV — case proposed by a Safran Electronics & Defense engineer | Gif-sur-Yvette, France

Project team

Nov 2025

- Identified the dynamics of a quadrotor delivered without technical documentation, from input-output measurements alone: data validation and outlier removal, sampling period justification, comparison of a first-order linear model against recurrent models (LSTM, RNN).
- Model Predictive Control built on the identified recurrent model, coupled with a Kalman filter for state estimation; identification in Python/PyTorch, weights exported to Simulink for closed-loop validation.

Lab project: Robot Fault Diagnosis through Domain Adaptation and Digital Twins — Industrial Engineering Laboratory, CentraleSupélec | 
France
DANN architecture and evaluation Jan 2025 – Jun 2025

- Since real fault data is scarce for robotic arms (built not to fail), classified real faults using a classifier trained solely on simulated data from a digital twin, with 90 real samples available in total.
- Designed a DANN (Domain-Adversarial Neural Network) on a 1-D convolutional extractor with residual blocks and squeeze-and-excitation attention: 87% accuracy on real data, against 86% for the client researcher’s existing MATLAB solution. Delivered a modular PyTorch implementation replacing the MATLAB prototype.

LEADERSHIP

Handi-tutorat (in partnership with Sopra Steria) **Gif-sur-Yvette, France**
Tutoring for students with disabilities Jan 2025 – Present

- Volunteered to support students with disabilities facing academic difficulties, helping them develop personalized learning methods and integrate into campus life.

Private Lessons **Paris Area, France**
Mathematics Tutor Sep 2024 – Jun 2025

- Provided academic support for high schoolers in mathematics, focusing on building their mathematical intuition and fostering independence.

SKILLS

Programming: Python (NumPy, Pandas, scikit-learn, PyTorch), SQL, OCaml, MATLAB, R;
Tools: Git, Bash, Jupyter, Slurm, HPC;
Languages: French (native), English (advanced), German (intermediate).

INTERESTS

Golf: Regular player;
Harp: 10 years, 7 years in conservatoire;
Associations: Cooking association of CentraleSupélec;